

The Fifth Workshop on Multiple Input Modalities and Sensations for VR/AR Interactions (MIMSVAI)

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Abstract

While virtual reality (VR) and augmented reality (AR) technologies have made rapid progress in recent years, their expansion is challenged by limited sensory feedback and input modalities. Natural interaction with digital environments depends upon the user's intuitive understanding of these environments. This type of understanding is built through user immersion, making realistic sensory feedback and innovative input mechanisms essential to engaging VR/AR experiences. For this, the fifth workshop on Multiple Input Modalities and Sensations for VR/AR Interactions, researchers in the ubiquitous computing (UbiComp) and VR/AR domains are invited to explore mechanisms to bring VR/AR interactions closer to those in the real world.

CCS Concepts

• Human-centered computing → Ubiquitous and mobile computing systems and tools.

Keywords

Virtual reality, Augmented reality, Mixed reality, Extended reality, Multiple sensory modalities, Multi-sensations, Mixed reality, Realism, User friendliness.

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1 Objective

As the immersive technologies of virtual reality (VR) and augmented reality (AR) develop, so has their popularity grown [7]. Although there have been several breakthrough developments in visual rendering and hardware capabilities, the user experience remains limited by input modalities and sensory feedback. In particular, developers are seeking means of seamlessly integrating multiple sensory feedback channels. User immersion in VR/AR environments can only be achieved when visual, auditory, haptic, and olfactory cues successfully converge [2].

However, ever-increasing realism in VR/AR environments depends not only on realistic sensory inputs, but also on their synthesis with interaction mechanisms. Users must feel that their digital interactions unfold naturally, through intuitive, unobtrusive interfaces. Developments in visual systems [3], haptic feedback [9], auditory cues [4], and even olfactory stimuli [6] have contributed to the enhancement of user experiences, as have the addition of diverse input modalities, including gaze tracking [5], voice commands [8], and gesture recognition [10]. While these contributions have moved virtual environments closer to the real world, much must be achieved before VR can truly replicate natural interactions.

To this end, the Multiple Input Modalities and Sensations for VR/AR Interactions (MIMSVAI 2025) workshop seeks to bring together leading researchers from ubiquitous computing (UbiComp)

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and VR/AR communities on October 12 or 13, 2025, in Espoo, Finland. This one-day event is part of the UbiComp/ISWC 2025 conference. It is designed as a catalyst for research into the immersive effects of VR/AR environments. It also aims to refine human-computer interactions by improving sensory integration and input modalities and sensations. With this in mind, MIMSAI 2025 calls for proposals of novel approaches to these functions. Papers accepted for this workshop will be presented in dedicated sessions, which will include moderated discussions. The workshop will include keynote talks by experts on VR/AR interactions. There will also be a sharing & discussion session where all participants will be able to share their research. The event is aimed at students and junior researchers, providing them with valuable learning and networking opportunities. Finally, a Best Paper Award will be presented with a certificate for outstanding research in this trending field.

2 Relevant and Timely

Ubiquitous computing (UbiComp) focuses on the seamless integration of computers into daily life. It is achieved through mobile and wearable devices, which allow human-computer interactions to unfold conveniently within our physical environment. Virtual reality (VR) seeks to immerse the user in a computer-generated environment and augmented reality (AR) overlays digital elements against physical surroundings. As technology progresses, the lines between these fields have begun to blur. UbiComp devices can now be used in VR/AR environments, and vice versa. Thus, the boundaries between physical and virtual experiences are starting to converge.

Our workshop aims to bring together researchers from both UbiComp and VR/AR fields to encourage the dynamic exchange of ideas and solutions in these fields. Our particular focus is the refinement of input modalities and sensory feedback to increase the degree of immersion achievable in VR/AR environments. This workshop offers a unified framework for these interdisciplinary areas. Together, we can create more engaging VR/AR user experiences and naturalized human-computer interactions.

3 Topics

Experts in UbiComp and VR/AR are invited to discuss several interrelated themes to achieve the aim of furthering collaboration on enhancing the user experience. A central theme is the development of intuitive input modalities to make user-computer interactions feel more natural. The pursuit of naturalness in human-computer interactions alongside the integration of realistic sensory feedback enhances the authenticity of virtual experiences. Another important aim is the development of devices and infrastructures to support immersive technologies and their ever-changing needs. New application scenarios for VR/AR will increase the scope of discussion, as participants reveal new opportunities for these technologies. Our discussion will also include novel design processes, platforms, and tools for immersive VR/AR environments. All of these themes are designed to support the integration of UbiComp and VR/AR innovations.



Figure 1: Image of the first keynote speaker at MIMSAI 2024.

- Innovative Inputs for Next-Generation Interactions.** The diversification of input modalities is crucial for the advancement of VR/AR. Systems must be able to accurately detect user actions and interpret their intentions. Human-computer interactions can be made more intuitive by either refining existing models or devising novel approaches that are more responsive.
- Pursuit of Realism through Sensory Feedback.** The authenticity of virtual worlds is key to user immersion. Sophisticated sensory feedback mechanisms are essential to the realism required for deep immersion. The integration of feedback mechanisms will allow users to engage both consciously and subconsciously, deepening their presence within the virtual world.
- Development of Next-Generation devices and Infrastructure.** Diverse input modalities and advanced sensory feedback will only be possible through improvements in devices and infrastructure. For technology to progress to the next generation, researchers must pioneer new VR/AR platforms and hardware to overcome current constraints.
- Identification of Emerging Applications** Expanding the scope of application for VR/AR technologies is crucial to its development. Current scenarios include gaming, healthcare, and education, but these fields have only scratched the surface of its potential. Our workshop is aimed at challenging reigning paradigms to reimagine the possibilities of these technologies.
- Integration with Generative AI** Generative AI creates digital content that is lifelike, engaging, and adaptive, greatly enhancing VR/AR environments. The fields of gaming, entertainment, healthcare, and education have been revolutionized by VR/AR applications; however, the untapped potential of this technology runs deep. Generative AI is poised to transform VR/AR environments to reach unprecedented heights of realism and interactivity. We invite visionary developers to contribute to this transformation with innovative ideas.



Figure 2: Image of participants of MIMSVAI 2024.

- **Establishing Libraries for Design Methodologies and Development Platforms and Tools.** Existing design processes, platforms, and toolkits must be adapted to evolving user requirements. Development environments such as Unity, Unreal Engine, Android ARCore, and Apple ARKit must be updated to support a continuous series of breakthroughs.

Our workshop is not strictly limited to these themes. Other significant topics within the VR/AR landscape will also be considered.

4 Workshop Structure

As suggested by the success of last year's event (see Figs. 1 and 2), MIMSVAI 2025 will be an in-person event running for one day. With an interactive agenda including participatory activities, keynote speeches, and interactive presentations aimed at fostering collaborative discussion and innovation (see Table 1). The workshop will open with a keynote presentation on VR/AR interactions, followed by a participant presentation and moderated discussion. The afternoon session will include another keynote presentation and second participant presentation. The final activity is designed to encourage engagement from students and junior researchers, who are welcome to share their thoughts with all participants. The concluding address will award a certificate to the authors of the Best Paper to recognize their outstanding research in the relevant topics.

We call for proposals from researchers in the fields of ubiquitous computing (UbiComp), human-computer interaction (HCI), VR/AR applications, haptic and tactile interfaces, and multi-sensory perception. All papers must fit the ACM sigconf template [1], which is limited to 5 pages (4 pages for content and 1 page for references). Rigorous review will then be performed by two Program Committee members and organization staff. Criteria include novelty, quality, and relevance. A high volume of submissions is anticipated and only about half will be accepted for presentation. Thus, a high quality of work is anticipated. Selected papers will be published in the UbiComp/ISWC Adjunct Proceedings and will be available through the ACM Digital Library.

Table 1: Workshop Schedule

Workshop Schedule	
9:00am-9:10am	Opening
9:10am-10:30am	Keynote Talk I: 50 + 20min
10:30am-10:50am	Coffee Break
10:50am-12:05pm	Paper Session I: $3 \times (20 + 5)$ min
12:05pm-2:00pm	Lunch Break
2:00pm-3:10pm	Keynote Talk II: 50 + 20min
3:10pm-3:30pm	Coffee Break
3:30pm-4:20pm	Paper Session II: $2 \times (20 + 5)$ min
4:20pm-4:50pm	Experience Sharing & Discussion Session
4:50pm-5:00pm	Closing & Best Paper Award Ceremony

5 Expected Outcomes

Our goal is to encourage collaboration and innovation in the relevant fields, enabling progress in interdisciplinary research. With an expected attendance of at least twenty participants, our workshop is likely to offer a comprehensive review of emerging trends in VR/AR and UbiComp.

Acknowledgments

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A Appendix: Call for papers

Recent developments in VR/AR are designed to increase user engagement and further increase the popularity of virtual environments. However, true realism remains out of reach, calling for the integration of novel input modalities and realistic sensory feedback. This workshop presents an opportunity for researchers in ubiquitous computing and VR/AR to collaborate on new methodologies to add depth and nuance to VR/AR worlds as these technologies approach new heights of realism.

Submitted papers may include, but are not limited to, the following topics:

- 2D/3D and volumetric display and projection technology
- Immersive analytics and visualization
- Modeling and simulation
- Multimodal capturing and reconstruction
- Scene description and management issues
- Storytelling
- Tracking and sensing
- Audio interfaces, sound rendering, spatialized audio, auditory perception, and psychoacoustics
- Embodied agents, virtual humans, and self-avatars
- Haptic and tactile interfaces, wearable haptics, passive haptics, pseudo haptics, and other touch-based user interfaces
- Mediated and diminished reality
- Multimodal input and output
- Multisensory rendering, registration, and synchronization
- Perception and cognition
- Presence, body ownership, and agency
- Teleoperation and telepresence
- 3D user interaction
- 3D user interface metaphors
- Collaborative interactions
- Ethical issues
- Human factors and ergonomics
- Input devices
- Locomotion and navigation

- Multimodal/cross-modal interaction and perception
- Non-fatiguing 3D user interfaces
- Non-visual interfaces (such as olfactory)
- Touch, tangible, and gesture interfaces
- Usage research, evaluation methods, and empirical studies
- Domain-specific VR/AR/MR/XR applications.
- Design processes, platforms/tools, or libraries used to incorporate input modalities or sensations for VR/AR interactions or study their use.
- The use of AI and Large Language Models in VR/AR/MR/XR applications

Important Dates

Paper submission: June 15, 2025

Paper notification: June 29, 2025

Camera-ready: July 30, 2025

Workshop: October 12/13, 2025

Best Paper Award

The Best Paper Award will recognize the most outstanding paper presented at the MIMSVAI 2025 workshop, celebrating excellence in research. The awardee will receive a certificate during the workshop.

Workshop Organizer

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Liwei Chan, National Yang Ming Chiao Tung University

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